

ViscoStar

Superior measurements of intrinsic viscosity
for macromolecular size and conformation



WYATT
TECHNOLOGY

ViscoStar®

Characterize polymer size, branching and more
with unmatched sensitivity

Benefits of SEC-IV

Intrinsic viscosity (IV) analysis is a versatile method for characterizing the physical properties of polymers and proteins. Although SEC-MALS is the gold standard for absolute molar mass and RMS radius, the characterization can be supplemented by the utilization of viscometry. Combined with size exclusion chromatography (SEC), also referred to as gel permeation chromatography (GPC), and a DAWN® or miniDAWN® multi-angle light scattering (MALS) detector, the ViscoStar determines:

- Molar mass and R_g
- Hydrodynamic radius R_h
- Intrinsic viscosity
- Conformation, branching ratios and drainage coefficients
- Mark-Houwink-Sakurada parameters relating the polymer's molar mass and intrinsic viscosity



SEC-MALS alone can determine the conformation and branching for distributions of RMS radius (R_g) greater than 10 nm; however, many polymers are either entirely below or have a least some fraction of the distribution below 10 nm. In this regime, viscometry is the natural solution to achieving branching and conformation analysis.

Intrinsic viscosity is also useful for characterizing polymers that are not amenable to MALS analyses, such as:

- Strongly fluorescent polymers when fluorescence is close to laser wavelength and cannot be blocked in MALS or when fluorescence-blocking filters are not installed

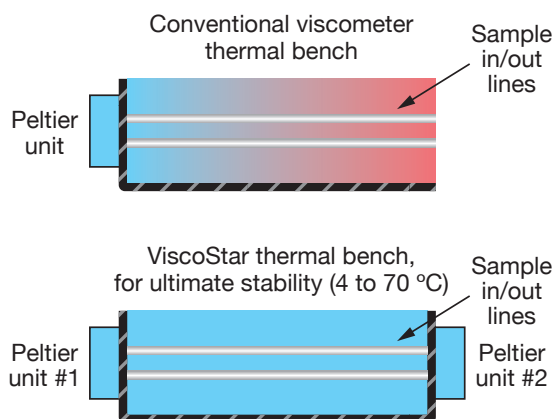
- Low dn/dc polymer/solvent systems that scatter light weakly
- Macromolecules that are dissolved in semi-opaque mobile phases or strongly absorb the MALS laser

Wyatt's ASTRA® software package for SEC supports both Universal Calibration and Mark-Houwink-Sakurada analyses. While these are not absolute methods like SEC-MALS, they are more robust than traditional column calibration techniques and require only the ViscoStar plus an online concentration detector. With the highest sensitivity, fastest response and best thermal stability, the ViscoStar is the premier differential viscometer for SEC and the ultimate companion for advanced SEC-MALS analyses.

Unique innovations

Thermal stability

The ViscoStar achieves highly uniform temperature across the entire fluidics path to ensure constant pressure differentials by means of an advanced thermal design, shown below.



A dual-Peltier design isolates the bridge from room-temperature variations.

Reliability

Multiple safeguards have been incorporated in the ViscoStar to enhance reliability and serviceability:

- Pressure transducers with all-316 stainless steel wetted surfaces for extended solvent compatibility and unparalleled corrosion-resistance.
- System pressure sensor to detect clogs and prevent transducer damage. With sensor logging, the system can determine when clogs are forming and recommend maintenance.
- Easy access to internal components

Thermal tuning

The resistive capillaries in a bridge must be finely balanced for optimal performance. Proprietary thermal bridge tuning dynamically ensures perfect balance in every run—with no moving parts.

How differential viscometry works

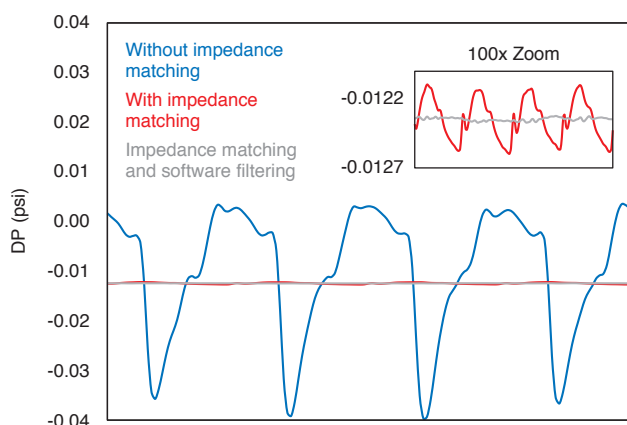
Differential viscometers measure very small changes in solution viscosity arising from the presence of macromolecules. This is accomplished with a balanced capillary bridge incorporating four capillaries of equal flow resistance R1-R4, shown here, and two pressure transducers IP and DP.

When pure solvent flows through all four capillaries, the bridge is balanced and DP reads zero. When sample is introduced, hold-up in the delay columns leads to an imbalance because sample flows through R1, R2 and R4 but solvent flows through R3.

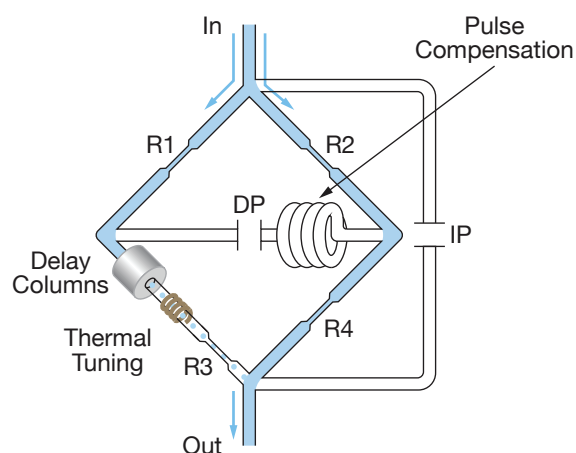
The viscosity difference between sample and solvent creates a pressure differential on DP, which can be analyzed to determine the specific viscosity $\eta_{sp}=4DP/(IP-2DP)$. The addition of an inline concentration detector (Optilab®) allows calculation of the sample's intrinsic viscosity.

Pulse compensation

Most differential viscometers suppress pressure fluctuations electronically—at the cost of chromatographic resolution. The ViscoStar introduces complete hardware-based impedance balancing of the bridge that minimizes pump pulses without impacting resolution. A proprietary algorithm eliminates residual pulse amplitude.



Patented pulse compensation technology eliminates pump fluctuations.



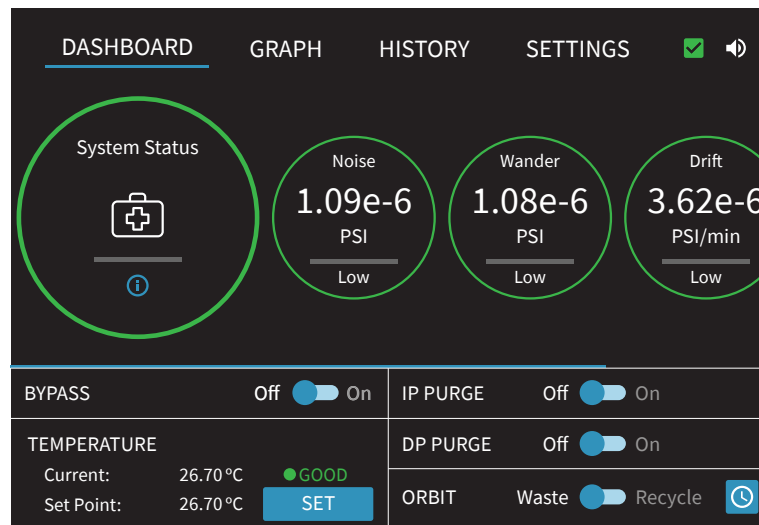
Without delay columns, the impedance of the capillary bridge would be fully balanced with the thermal tuning element alone. The pulse compensation element matches the additional dynamic impedance of the delay columns, eliminating the effect of pump pulses on the DP transducer.

New User Interface

The larger responsive display is the starting point of our Smart Services™ platform and designed to give the user a simple to use, all-in-one system view

SEC-MALS
System
Readiness
Monitor

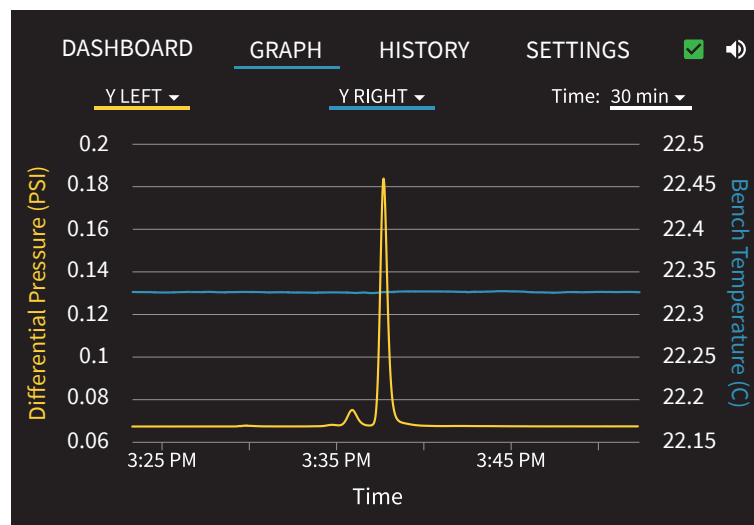
Customizable
User
Preferences



Real-time
Instrument
Health
Indicators

System
Control
Panel

Zoom,
Pinch,
Swipe
Functionality



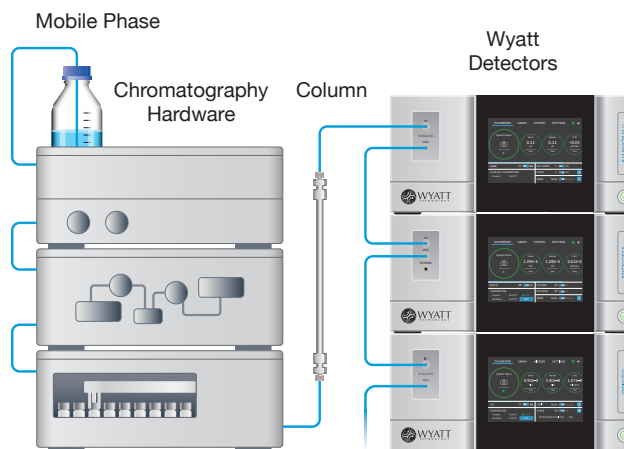
Customizable
Graphing

Smart Services Platform

A series of user experience enhancements
delivering greater ease-of-use and improved workflow

System Ready Monitor

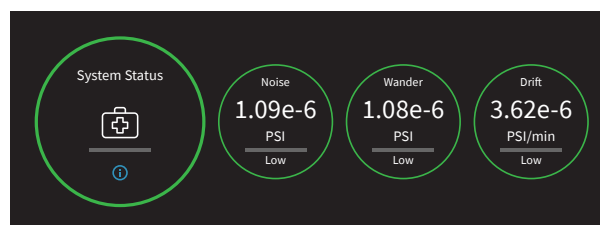
Never waste a run due to incomplete equilibration, excessive mobile-phase noise or sub-optimal detector state. The System Ready Monitor simply does that for you and continuously reports if all systems are optimal right from the front panel. If problems do arise, the System Ready Monitor alerts you by changing from green to yellow to red depending on the severity and provides actionable, real-time guidance on what needs to be done to bring the system back to peak health.



Real-Time Health Indicators

System Ready Monitor indicating that not all is well? For more detailed information on what's holding you back, review the Real-Time Health indicators. Perhaps the Forward Monitor indicator is triggered by a bubble or the Drift indicator by insufficient column equilibration.

Specific indicators can be customized for more or less stringent requirements. For example, the Noise indicator can be set to have a wider acceptable range for aqueous buffers, which typically exhibit more noise than organic solvents.

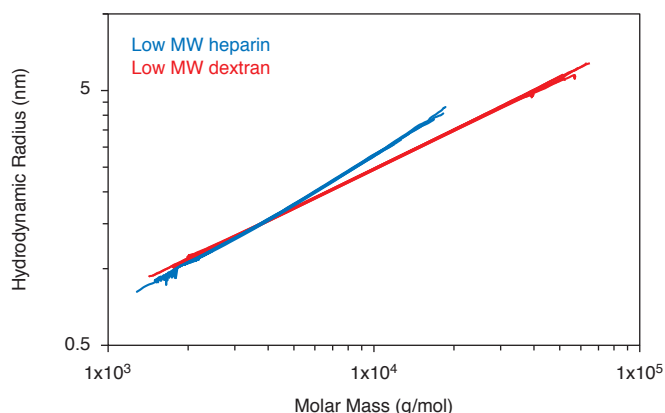


CheckPlus™

If concerns about the status of your instrument arise, an expert opinion is just a click away. With CheckPlus, our integrated service application, a complete instrument and system history report can be auto-generated at any time and sent to a Wyatt Service Engineer or Application Scientist for further analysis. This is just another example of how our Smart Services Platform is designed to deliver important customer-facing benefits and simplified use.

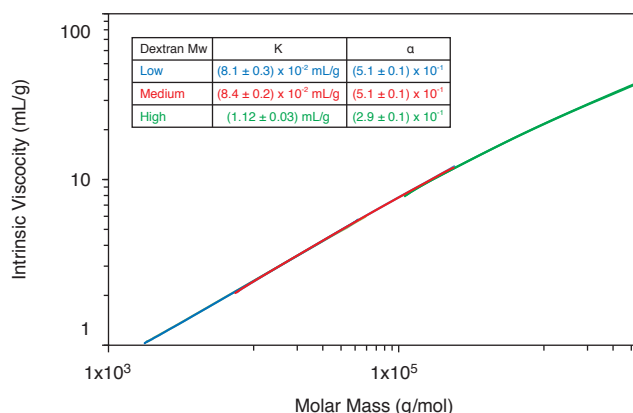


Small polymer sizing



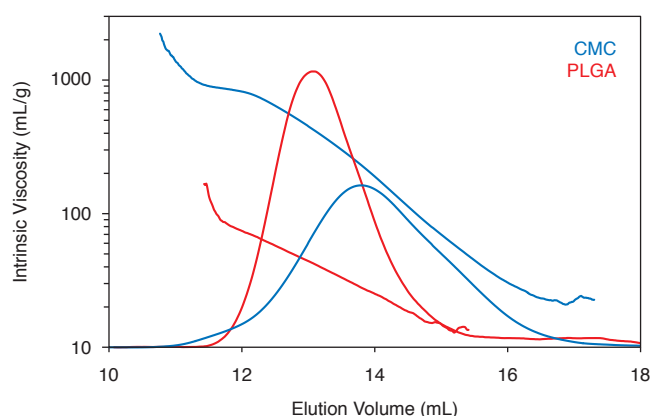
For a polydisperse polymer with an RMS radius less than 10 nm (or molar mass less than 100 kDa), viscometry is the best choice for measuring size and conformation. Here, the hydrodynamic radius of a low MW heparin and low MW dextran are plotted against molar mass determined with DAWN MALS and Optilab dRI detectors. Each sample was injected twice under four different SEC conditions, including different columns, mobile phases and flow rates. The results demonstrate outstanding reproducibility and consistency of our triple detection SEC system.

Conformational change with MW



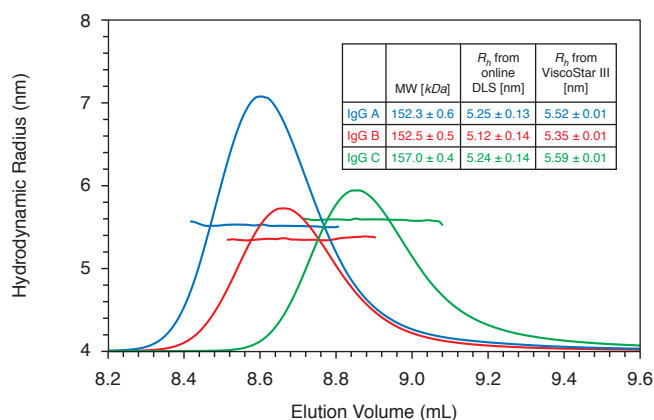
A Mark-Houwink-Sakurada (MHS) plot shows intrinsic viscosity as a function of molar mass—revealing the polymer conformation. The MHS plots of low, medium and high MW dextrans, shown here, indicate conformational change with increasing molar mass of the molecules. When a linear model is used as the reference, ASTRA further quantifies branching ratios and branch units-per-molecule for a variety of branching models, including comb and star polymers.

Intrinsic viscosity across a wide range



The ViscoStar provides the highest available dynamic range and sensitivity in online viscometry. This graph shows the intrinsic viscosities of 1) poly(lactic co-glycolic acid), or PLGA, in THF and 2) carboxymethyl cellulose, or CMC, in aqueous mobile phase, both measured with a ViscoStar and Optilab, spanning a range from tens to thousands of mL/g. In other examples (on this page), the same instruments readily measure intrinsic viscosity values in the low single digits with excellent signal-to-noise, even at very low concentrations.

Precise R_h measurement of proteins



Three IgG proteins with similar MWs eluted at different elution volumes. To understand their different elution properties, online DLS is typically used to measure the hydrodynamic radius, R_h , of a protein. However, online DLS did not provide reliable insights, in this case due to the relatively high experimental uncertainty of the DLS from low concentration. The precise R_h values obtained with a MALS detector and the ViscoStar, on the other hand, clearly revealed the causes. IgG B eluted later than IgG A due to its more compact structure; IgG C had a slight interaction with the SEC column.

Specifications

Differential Viscosity

Sensitivity (minimum detectable sample)	0.1 µg of 100 kDa polystyrene in THF at 1 mL/min, under typical GPC conditions
Differential Pressure Noise	< 0.05 Pa
Differential Pressure Range	6.9 kPa, protected against overpressure
Dynamic Range	135,000:1
Linearity	0.5% of full scale
Differential Pressure Drift	< 2.5 Pa/hr
Capillary Bridge	Thermal leak-free mechanism
Autobalance	
Sample Shear Rate	5000 Hz with water at 1 mL/min

Fluidics

Maximum Flow Rate	Aqueous: 1.5 mL/min; THF: 3 mL/min
Delay Volume Options	16.2, 10.8 or 5.4 mL, with additional factory-installed option for 32.4 mL*
Maximum Permissible Backpressure at Inlet	690 kPa
Wetted Materials	316 stainless steel, PTFE, zirconium silicate, Kalrez®, Inconel, 26-1 Stainless Steel

Temperature Control

Range	4 °C to 70 °C
Stability	< 0.005 °C
Ambient Range	0 °C to 40 °C, 20% to 70% RH

Electronics

Digital Output	20 bits @ 10 Hz
Auxiliary I/O	
Inputs	Autoinject contact closure, Alarm In, Analog In: 2 differential inputs, 16 bits resolution, range -10 V to +10 V
Outputs	Autoinject Retransmit, Alarm Out, Recycle Out
On-board Computer	
Communications	Ethernet

Dimensions	58 cm (L) x 36 cm (W) x 18 cm (H)
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*Volume of fluid through the ViscoStar until the retained sample elutes. Actual volume of fluid in the column is half this value. Wyatt Technology is committed to continual improvement. Specifications subject to change without notice.

Warranty: All Wyatt instruments are guaranteed against manufacturing defects for 1 year.

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Left to Right

Geoffrey K. Wyatt, Chief Executive Officer
Dr. Philip J. Wyatt, Chairman of the Board
Clifford D. Wyatt, President

For more than 35 years, we've operated as one of the very few remaining family-owned businesses in the analytical instrument industry. With installations in more than 65 countries, over 15,000 refereed journal publications citing our instruments and more than 25 PhD scientists on staff, we take great pride in the worldwide recognition that Wyatt Technology has received as a leading manufacturer of instruments and software for absolute macromolecular and nanoparticle characterization. Our dedication to providing customers with comprehensive training and personal support has made us the gold standard in this field.

The ViscoStar is one of many tools in Wyatt's Light Scattering Toolkit for Essential Macromolecular and Nanoparticle Characterization.

Learn more at www.wyatt.com

